

MongoDB Concepts Explained



You said:

mangodb all concept explain

ChatGPT said:

Here's a **complete MongoDB guide (beginner → advanced)** explained in a simple, interview-ready way. This will cover everything you need for **MERN / backend / Cuvette prep**.

1. What is MongoDB?

MongoDB is a **NoSQL database** that stores data in **JSON-like documents (BSON)** instead of tables.

 Example:

2. MongoDB vs SQL

Feature	MongoDB	SQL
Structure	Document (JSON)	Table (Rows/Columns)
Schema	Flexible	Fixed
Scaling	Horizontal	Vertical

Feature	MongoDB	SQL
Relations	Embedded / Reference	Joins

3. Basic Structure

SQL	MongoDB
Database	Database
Table	Collection
Row	Document

👉 Example:

4. CRUD Operations

➤ CREATE

➤ READ

➤ UPDATE

➤ DELETE

5. Query Operators

► Comparison

Operator	Meaning
\$gt	greater than
\$lt	less than
\$gte	> =
\$lte	< =
\$ne	not equal

► Logical

Operator	Meaning
\$and	AND
\$or	OR
\$not	NOT

6. Projection (Select Fields)

👉 Only returns **name**

7. Sorting, Limit, Skip

8. Update Operators

👉 Example:

9. Delete Operations

10. Indexing (IMPORTANT FOR INTERVIEW)

👉 Improves query performance

✓ Faster search ✗ Takes extra memory

11. Aggregation Pipeline (VERY IMPORTANT)

Like SQL GROUP BY

👉 Stages:

- \$match → filter
- \$group → group data
- \$sort → sort
- \$project → reshape

12. Schema Design

➤ **Embedded (Best for small data)**

► Referenced (Large data)

13. Relationships

Type	When to Use
Embedded	Small, frequently accessed
Reference	Large, reusable data

14. Mongoose (MongoDB with Node.js)

Mongoose is an ODM (Object Data Modeling library)

► Schema

► Model

15. Validation

16. Population (JOIN in MongoDB)

17. Middleware (Hooks)

18. Authentication (IMPORTANT)

Use:

- JWT

- bcrypt

👉 Example:

19. Transactions

Used for:

- Banking apps
- Payments

20. Performance Optimization

✔ Use indexing ✔ Avoid large documents ✔ Use projection ✔ Use aggregation wisely

❓ 21. Common Interview Questions

Q1: What is MongoDB?

👉 NoSQL database storing JSON-like documents

Q2: Difference between MongoDB & SQL?

👉 Schema vs No schema

Q3: What is aggregation?

👉 Data processing pipeline

Q4: What is indexing?

👉 Improves query speed

22. Real-World Example (Backend)

Final Roadmap

1. Basics (CRUD)

2. Queries
3. Aggregation
4. Schema Design
5. Mongoose
6. Authentication
7. Performance

If you want next level 🚀 👉 I can give you:

- 50 MongoDB interview questions
- Full CRUD backend project
- MongoDB + Express + React complete project

Just tell me 👉

2/2

You said:

The mango shell explain all concept

ChatGPT said:

Got it—you mean **MongoDB Shell (mongosh)**, not “mango” 😊. Let’s break down **all important MongoDB shell concepts** in a clean, interview-ready way.

What is MongoDB Shell?

MongoDB Shell (mongosh) is an **interactive command-line tool** used to:

- Connect to MongoDB
- Run queries
- Perform CRUD operations
- Debug & test database logic

👉 It uses **JavaScript syntax**, so if you know JS, it feels natural.



1. Start Mongo Shell

👉 Connect to local DB:



2. Database Operations

➤ Show all databases

➤ Create / Switch DB

👉 If DB doesn't exist → it will be created automatically after inserting data.

➤ Current DB



3. Collections

➤ Show collections

➤ Create collection



4. CRUD in Shell

➤ CREATE

► **READ**

👉 Pretty print:

► **UPDATE**

► **DELETE**

 5. Query Operators

► **Comparison**

Operator	Meaning
\$gt	>
\$lt	<
\$gte	> =
\$lte	< =
\$ne	!=

► **Logical**

 6. Projection

👉 Only returns **name**

▼ 7. Sorting / Limit / Skip

🔧 8. Update Operators

👉 Example:

⚡ 9. Indexing (Important)

👉 Benefits:

- Faster queries
- Better performance

💡 10. Aggregation Pipeline

👉 Stages:

- **\$match** → filter
- **\$group** → group data
- **\$sort** → sort
- **\$project** → reshape

🔗 11. Relationships

► Embedded

► Referenced

12. Useful Shell Commands

► Drop collection

► Drop database

► Count documents

13. Help Commands

14. JavaScript in Shell

👉 You can run JS directly:

15. Authentication (Shell)

16. Tips for Interview

✔ mongosh uses JavaScript ✔ No fixed schema ✔ Use indexing for performance ✔ Aggregation is very important ✔ CRUD must be strong

Quick Revision

👉 In shell you can:

- Create DB & collections

- Insert / Read / Update / Delete
- Run queries
- Use aggregation
- Manage indexes

If you want next level 🚀 I can give you:

- 🔥 50 MongoDB shell practice questions
- 🧠 Real interview coding questions
- 💻 Full backend project using mongosh + Node

Just tell me 👍

2/2

You said:

how we store in data? (BSON)

ChatGPT said:

🚀 How MongoDB Stores Data (BSON Explained Simply)

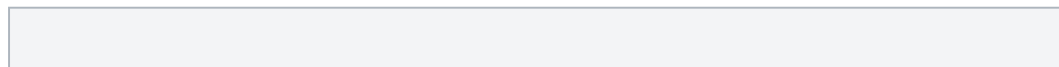
MongoDB stores data in **BSON** format.

🧠 What is BSON?

BSON = Binary JSON

👉 It is a **binary-encoded version of JSON** used internally by MongoDB.

JSON Example:



BSON stores the same data in binary format for:

- Faster reading
- Faster writing

- Better indexing
- More data types support

 Why BSON Instead of Normal JSON?

Feature	JSON	BSON
Human readable	Yes	No
Binary format	No	Yes
Speed	Slower	Faster
Extra types	Limited	More
Date support	Weak	Strong

 BSON Supports Extra Data Types

MongoDB can store:

- String
- Number
- Boolean
- Array
- Object
- Date
- ObjectId
- Null
- Binary Data

 Example Document



How Data is Organized

Structure:

Example:



Storage Process

Step 1:

You insert data:

Step 2:

MongoDB converts JSON → BSON

Step 3:

Stores it on disk efficiently



_id Field

Every document gets a unique ID automatically:

👉 Used for:

- Unique identification
- Fast search
- Primary key



BSON Advantages

✓ Compact ✓ Fast ✓ Efficient ✓ Supports rich data types ✓ Better for large applications

BSON Limitations

- Slightly larger than JSON
- Not human-readable
- Needs parsing

Real Example

Stored internally as BSON.

BSON vs SQL Storage

SQL:

MongoDB:

Interview Answer

Q: How does MongoDB store data?

 MongoDB stores data in BSON (Binary JSON) documents inside collections.

Simple Memory Trick:

JSON = You write

BSON = MongoDB stores

Final Summary

MongoDB:

- Accepts JSON-like documents
- Converts to BSON

- Stores in collections
- Uses ObjectId
- Provides fast access

If you'd like, I can also explain:

- BSON vs JSON deeply
- ObjectId structure
- Internal storage engine (WiredTiger)
- Real-world schema examples

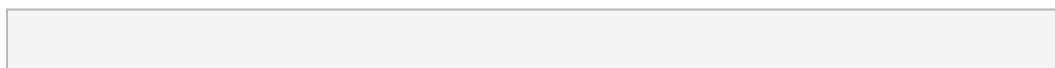
You said:

ChatGPT said:

MongoDB Document and Collection (Easy Explanation)

In **MongoDB**, data is stored differently than SQL.

Basic Structure

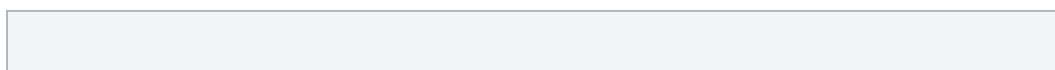


1. What is a Document?

A **Document** is a single record (like one row in SQL).

👉 Stored in **BSON** format.

Example:



SQL Comparison:

Features of Document:

✓ JSON-like structure ✓ Flexible fields ✓ Can store nested data ✓ Unique `_id` automatically

Example with `_id`:

2. What is a Collection?

A **Collection** is a group of multiple documents.

👉 Similar to a table in SQL.

Example:

SQL Comparison:

SQL MongoDB

Table Collection

Real Example

Create Collection:

Insert Documents:

Flexible Schema Advantage

Different documents in the same collection can have different fields.

👉 SQL usually requires fixed columns.

Common Commands

Show collections:

Find documents:

Insert one document:

Nested Documents Example

Key Differences

Feature	Document	Collection
Meaning	Single record	Group of records
SQL Equivalent	Row	Table
Format	BSON	Multiple BSON docs

Interview Questions

Q: What is a document?

👉 A BSON record storing data in key-value format.

Q: What is a collection?

👉 A group of MongoDB documents.



Easy Memory Trick

Document = One student form

Collection = Full classroom data



Final Summary

✓ Document = Single data entry ✓ Collection = Group of documents ✓
Flexible schema ✓ Stored in BSON

Example:

You said:

ChatGPT said:



Insert Data in MongoDB (mongosh)

In **MongoDB**, inserting means adding documents into a collection.



Basic Syntax

➤ Insert One Document



Example:



Output:

Insert Multiple Documents

► Syntax:

Example:

View Inserted Data

Important Points

✓ If collection doesn't exist → MongoDB creates it automatically ✓ `_id` field is generated automatically ✓ Data stored as BSON

Custom `_id`

Duplicate `_id` Error

MongoDB won't allow duplicate `_id`.

Insert Data Types Example

SQL Comparison

SQL:

MongoDB:

Practice Commands

Insert Student:

Insert Many Products:

Interview Questions

Q: Difference between insertOne and insertMany?

- 👉 `insertOne()` inserts one document
- 👉 `insertMany()` inserts multiple documents

Easy Memory Trick

`insertOne` → One document

`insertMany` → Many documents

Final Summary

- ✓ `insertOne()` → Single insert
- ✓ `insertMany()` → Bulk insert
- ✓ Auto creates collection
- ✓ BSON storage
- ✓ `_id` auto-generated

Example:

You said:

ChatGPT said:



show collections in MongoDB Shell

`show collections` is used to display **all collections (tables)** inside the current database.



Syntax



Example

Step 1: Switch to database

Step 2: Show collections



Output:



Meaning:

Your database contains:

- `students` collection
- `teachers` collection

- **courses** collection

SQL Comparison

SQL

MongoDB

SHOW TABLES; show collections

Important Notes

✓ Shows collections only in the current database ✓ If no collections exist → nothing displayed ✓ Useful for checking database structure

Create Collection Example

Now:

Output:

Related Commands

Show databases:

Current database:

Interview Question

Q: How do you list all collections in MongoDB?

 Use:

Easy Memory Trick

Database = Folder

Collection = Files inside folder

👉 `show collections` = Show all files

Final Summary

✓ Displays all collections ✓ Similar to SQL tables ✓ Helps inspect DB structure

Command:

You said:

ChatGPT said:

`insertMany()` in MongoDB

`insertMany()` is used to insert **multiple documents at once** into a collection.

Syntax

Example

Output:

View Inserted Data

Benefits of `insertMany()`

✓ Faster bulk insertion ✓ Saves time ✓ Better performance ✓ Useful for large datasets

Auto Collection Creation

If `users` collection doesn't exist, MongoDB creates it automatically.

Ordered vs Unordered Insert

Default (Ordered = true)

Stops if one document fails:

Unordered

Continues even if one fails:

Duplicate `_id` Example

👉 Error because `_id` must be unique.

SQL Comparison

SQL:

MongoDB:

Practice Example

Interview Questions

Q: What is insertMany?

👉 Used to insert multiple documents in one command.

Q: Why use insertMany?

👉 Better speed and bulk operations.

Easy Memory Trick

insertOne → **One** document

insertMany → **Many** documents

Final Summary

✓ Bulk insert ✓ Faster than multiple insertOne ✓ Auto `_id` generation ✓
Better for large data

Final Command:

You said:

ChatGPT said:

`find()` in MongoDB (Search Data)

In **MongoDB**, `find()` is used to **read/search documents** from a collection.



Basic Syntax

- **query** → filter data (optional)
- **projection** → select fields (optional)



1. Get All Documents



Returns all data in `users`



2. Find Specific Data



Returns documents where `name = Amir`



3. Find One Document



Returns only the first matching document



4. Pretty Output



Makes output readable



5. Query Operators

► Greater than



► Less than

► AND condition

► OR condition



6. Projection (Select Fields)

👉 Shows only **name**



7. Sort Data



8. Limit & Skip



9. Nested Field Search



10. Count Documents



SQL Comparison

SQL:

MongoDB:

Practice Examples

Interview Questions

Q: Difference between find() and findOne()?

👉 `find()` → returns multiple documents

👉 `findOne()` → returns single document

Easy Memory Trick

find() → Find many

findOne() → Find one

Final Summary

✓ Used to read data ✓ Supports filters ✓ Supports projection ✓ Supports sorting & pagination

Final Example:

2/2

You said:

FIND in DB Query Operators Q. Find students where marks > 75
`db.student.find( {marks: {$gt: 75}} )` Q. Find students who live in Delhi or Mumbai
`db.student.find( {city: {$in: ["Delhi", "Mumbai"]}} )` Q. Find students who scored > 75 or live in Delhi
`db.student.find( { $or: [ {marks: {$gt: 75}}, {city: "Delhi"} ] } )`

ChatGPT said:

MongoDB `find()` with Query Operators (Student Examples)

Here's the corrected and easy explanation for your MongoDB queries.

1. Find students where marks > 75

 **Command:**

 **Meaning:**

- `marks` → field name
- `$gt` → greater than

 Returns all students with marks above 75.

2. Find students who live in Delhi or Mumbai

 **Command:**

 **Meaning:**

- `$in` → matches any value in the array

 Returns students from Delhi OR Mumbai.

3. Find students who scored > 75 OR live in Delhi

 **Command:**

Meaning:

- `$or` → either condition can be true

 Returns:

- Students with marks > 75 **OR**
- Students living in Delhi

Sample Data

Other Useful Query Operators

Operator	Meaning
<code>\$gt</code>	Greater than
<code>\$lt</code>	Less than
<code>\$gte</code>	Greater than or equal
<code>\$lte</code>	Less than or equal
<code>\$in</code>	Match values in list
<code>\$or</code>	OR condition
<code>\$and</code>	AND condition
<code>\$ne</code>	Not equal

More Practice Examples

Marks between 70 and 90:

Students not in Delhi:

SQL Comparison

SQL:

MongoDB:

Interview Tip

Q: Difference between \$in and \$or?

👉 \$in checks one field for multiple values

👉 \$or checks multiple conditions

Easy Memory Trick

\$gt → **Greater Than**

\$in → **Inside list**

\$or → **Any one condition**

Final Summary

✓ find() = Search documents

✓ \$gt = marks > value

✓ \$in = multiple city match

✓ \$or = combine conditions

You said:

ChatGPT said:

MongoDB Query Operators (Complete Guide)

Query Operators in **MongoDB** are special commands used inside `find()` to filter data.

👉 They help you search documents based on conditions.

Basic Syntax

1. Comparison Operators

These compare values.

➤ **\$gt (Greater Than)**

👉 Marks > 75

➤ **\$lt (Less Than)**

➤ **\$gte (Greater Than or Equal)**

➤ **\$lte (Less Than or Equal)**

➤ **\$ne (Not Equal)**

2. Logical Operators

➤ **\$and**

➤ **\$or**

➤ **\$not**

3. Array Operators

➤ **\$in**

👉 City is Delhi or Mumbai

➤ **\$nin**

👉 City is NOT Delhi/Mumbai

4. Element Operators

➤ **\$exists**

👉 Documents where age field exists

5. Evaluation Operators

➤ **\$regex**

👉 Names starting with A

Practical Examples

Students with marks between 70–90:

Students in Delhi with marks > 75:

SQL Comparison

SQL:

MongoDB:

Most Important Operators for Interview

Operator	Meaning
\$gt	Greater than
\$lt	Less than
\$gte	> =
\$lte	< =
\$ne	Not equal
\$in	In list
\$nin	Not in list
\$or	OR
\$and	AND
\$exists	Field exists
\$regex	Pattern search

Interview Questions

Q: What is **\$in**?

 Matches values from multiple options.

Q: Difference between \$or and \$in?

- 👉 \$or checks multiple conditions
- 👉 \$in checks one field against many values

🚀 Easy Memory Trick

\$gt → Greater Than

\$lt → Less Than

\$in → Inside list

\$or → Any condition

🎯 Final Summary

- ✔ Query operators filter data
- ✔ Used inside find()
- ✔ Comparison + logical + array + regex
- ✔ Essential for backend interviews

You said:

ChatGPT said:

🚀 MongoDB Update Operations (Update Data in DB)

In **MongoDB**, update commands are used to modify existing documents.

🧱 Main Update Methods

Method	Purpose
updateOne()	Update first matching document
updateMany()	Update multiple documents

Method	Purpose
--------	---------

<code>replaceOne()</code>	Replace full document
---------------------------	-----------------------



1. `updateOne()`

Syntax:

Example:



Meaning:

- Find student named Aamir
- Update marks to 95



2. `updateMany()`

Example:



Updates all matching students.



3. `replaceOne()`



Replaces entire document except `_id`



Important Update Operators



`$set`

Update specific field:

➤ **\$inc**

Increase numeric value:

➤ **\$unset**

Remove field:

➤ **\$push**

Add item to array:

➤ **\$pull**

Remove item from array:

 Upsert Option

If document doesn't exist → create it.

 SQL Comparison

SQL:

MongoDB:

Practice Examples

Increase marks:

Add course:

Interview Questions

Q: Difference between `updateOne` and `updateMany`?

- 👉 `updateOne()` updates first match
- 👉 `updateMany()` updates all matches

Q: What is `$set`?

- 👉 Updates or adds field value

Easy Memory Trick

`$set` → Set value

`$inc` → Increase

`$push` → Add

`$pull` → Remove

Final Summary

- ✓ `updateOne()` → single update
- ✓ `updateMany()` → bulk update
- ✓ `$set`, `$inc`, `$push`, `$pull` are key
- ✓ `upsert` creates if missing

You said:

ChatGPT said:



Nesting in MongoDB (Nested Documents)

In **MongoDB**, **nesting** means storing objects or arrays inside a document.



This allows complex data to be stored in one document.



Basic Example



Explanation:

- **name** → normal field
- **address** → nested object



Why Use Nesting?

✓ Related data stays together ✓ Faster reads ✓ Fewer joins ✓ Better for user profiles, orders, settings



Insert Nested Document



Find Nested Data

Find by nested field:



Dot notation:



Update Nested Field

Nested Arrays Example

Add new skill:

Deep Nesting Example

Query:

Embedded vs Referenced

Type	Meaning
Embedded/Nested	Data inside document
Referenced	Linked by ID

Embedded Example:

Advantages

✓ Better performance ✓ Easy retrieval ✓ Real-world modeling

Disadvantages

✗ Large document size ✗ Harder updates if too deep ✗ Duplication possible

Interview Questions

Q: What is nesting?

👉 Storing objects/arrays inside documents.

Q: How to query nested fields?

👉 Use dot notation.

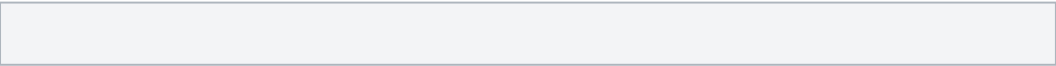
 Easy Memory Trick

Nested = Document inside document

 Final Summary

✔ Nested documents store related data together ✔ Use dot notation for search/update ✔ Great for profiles, addresses, orders

Example:



You said:

ChatGPT said:

 MongoDB Delete Operations (Delete Data from DB)

In **MongoDB**, delete commands remove documents from a collection.

 Main Delete Methods

Method	Purpose
<code>deleteOne()</code>	Delete first matching document
<code>deleteMany()</code>	Delete multiple matching documents
<code>drop()</code>	Delete entire collection

Method	Purpose
--------	---------

<code>dropDatabase()</code>	Delete whole database
-----------------------------	-----------------------



1. `deleteOne()`

Syntax:

Example:



Meaning:

Deletes the first student where `name = Aamir`



2. `deleteMany()`

Example:



Deletes all students from Delhi.



3. Delete All Documents



Empty filter `{}` means all documents.



Important:

This removes data but keeps the collection.



4. Drop Entire Collection

👉 Deletes:

- Collection
- All documents
- Indexes

5. Drop Entire Database

👉 Deletes complete current database.

SQL Comparison

SQL:

MongoDB:

Practice Examples

Delete students with marks < 40:

Delete all products:

Return Output Example

Important Tips

- ✓ Always check filter before delete ✓ `deleteMany({})` removes all documents
- ✓ `drop()` removes collection
- ✓ `dropDatabase()` removes DB

Interview Questions

Q: Difference between `deleteOne` and `deleteMany`?

- 👉 `deleteOne()` deletes first match
- 👉 `deleteMany()` deletes all matches

Q: Difference between `deleteMany({})` and `drop()`?

- 👉 `deleteMany({})` keeps collection
- 👉 `drop()` removes collection completely

Easy Memory Trick

`deleteOne` → One

`deleteMany` → Many

`drop` → Full collection

Final Summary

- ✓ `deleteOne()` → single delete
- ✓ `deleteMany()` → bulk delete
- ✓ `drop()` → collection delete
- ✓ `dropDatabase()` → full DB delete

Final Example:

You said:

mongodb all concept explain one by one easy way to understand

ChatGPT said:



Complete MongoDB Guide (Beginner → Advanced)

This guide explains **MongoDB** step by step in an easy way for interviews, backend development, MERN stack, and practical projects.



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1 Introduction to MongoDB

MongoDB is a NoSQL database that stores data in flexible JSON-like documents.

Example:

Key Features:

- No fixed schema
- High performance
- Scalable
- JSON/BSON format
- Great for MERN stack

2 MongoDB vs SQL

Feature	MongoDB	SQL
Data Format	Document	Table
Schema	Flexible	Fixed
Scaling	Horizontal	Vertical

Relations Embedded/Referenced Joins

3 Installation & Setup

Install:

- MongoDB Community Server
- MongoDB Compass
- Mongo Shell (**mongosh**)

Start Shell:

4 MongoDB Shell Basics

Show databases:

Use database:

Show collections:

5 Databases, Collections, Documents

Structure:

SQL Comparison:

- Database = Database
- Table = Collection
- Row = Document

6 BSON Format

MongoDB stores data as **BSON (Binary JSON)**.

Benefits:

- Faster storage
- Supports extra data types
- Better indexing

- Efficient retrieval

7 **CRUD Operations**

CREATE

Insert One:

Insert Many:

READ

Find all:

Find specific:

UPDATE

Update one:

DELETE

Delete one:

8 **Query Operators**

Operator	Meaning
\$gt	Greater than
\$lt	Less than
\$gte	Greater than equal
\$lte	Less than equal
\$ne	Not equal
\$in	Match list
\$or	OR condition

\$and AND condition

Example:

9 Projection

Select specific fields:

10 Sorting, Limiting, Skipping

Sort:

Limit:

Skip:

1 1 Update Operators

Operator	Purpose
\$set	Update value
\$inc	Increment
\$push	Add to array
\$pull	Remove from array

\$unset Remove field

1 2 Delete Operations

Delete many:

Drop collection:

1 3 Nested Documents

Query nested:

1 4 Arrays in MongoDB

Add item:

Remove item:

1 5 Indexing

Create index:

Benefits:

- Faster search
- Better query performance

1 6 Aggregation Pipeline

Example:

Stages:

- \$match
- \$group
- \$sort
- \$project

- \$limit

1 7 Schema Design

Embedded:

Referenced:

1 8 Relationships

Embedded:

- Fast reads
- Small related data

Referenced:

- Large reusable data
- Better normalization

1 9 Mongoose Basics

Schema:

Model:

2 0 Validation

2 1 Population

Purpose:

MongoDB version of JOIN.

2 2 Middleware

Pre-save hook:

2 3 Authentication

Password hashing:

Token:

2 4 Transactions

Used for:

- Payments
- Banking
- Multi-step secure operations

2 5 Performance Optimization

Best Practices:

- Use indexes
- Use projection
- Avoid deep nesting
- Optimize aggregation
- Limit returned fields

2 6 Common Interview Questions

Q1: What is MongoDB?

A NoSQL document database.

Q2: What is BSON?

Binary JSON storage format.

Q3: Difference between SQL and MongoDB?

Tables vs documents.

Q4: What is indexing?

Improves search speed.

Q5: What is aggregation?

Data processing pipeline.

Real-World Backend Example

Final Learning Roadmap

Beginner:

- MongoDB basics
- CRUD
- Query operators
- Nesting

Intermediate:

- Indexing
- Aggregation
- Schema design
- Relationships

Advanced:

- Mongoose
- Authentication
- Transactions

- Optimization

Easy Memory Trick

Database → Collection → Document

JSON input → BSON storage

CRUD → Core operations

Aggregation → Advanced analysis

Final Summary

MongoDB is:

- Flexible
- Powerful
- Scalable
- MERN-friendly
- Interview essential

Master these concepts one by one, practice commands daily, and you'll be strong in backend + full-stack interviews.

I created a complete step-by-step MongoDB guide in the canvas covering all major concepts from beginner to advanced, including:

- MongoDB basics
- BSON
- Documents & Collections
- CRUD operations
- Query operators
- Projection
- Sorting / Limit / Skip
- Update & Delete

- Nesting
- Arrays
- Indexing
- Aggregation pipeline
- Schema design
- Relationships
- Mongoose
- Validation
- Population
- Middleware
- Authentication
- Transactions
- Performance optimization
- Interview questions
- Real-world backend examples

It's organized for easy learning, revision, and interview preparation.